

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{EE} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{EE} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

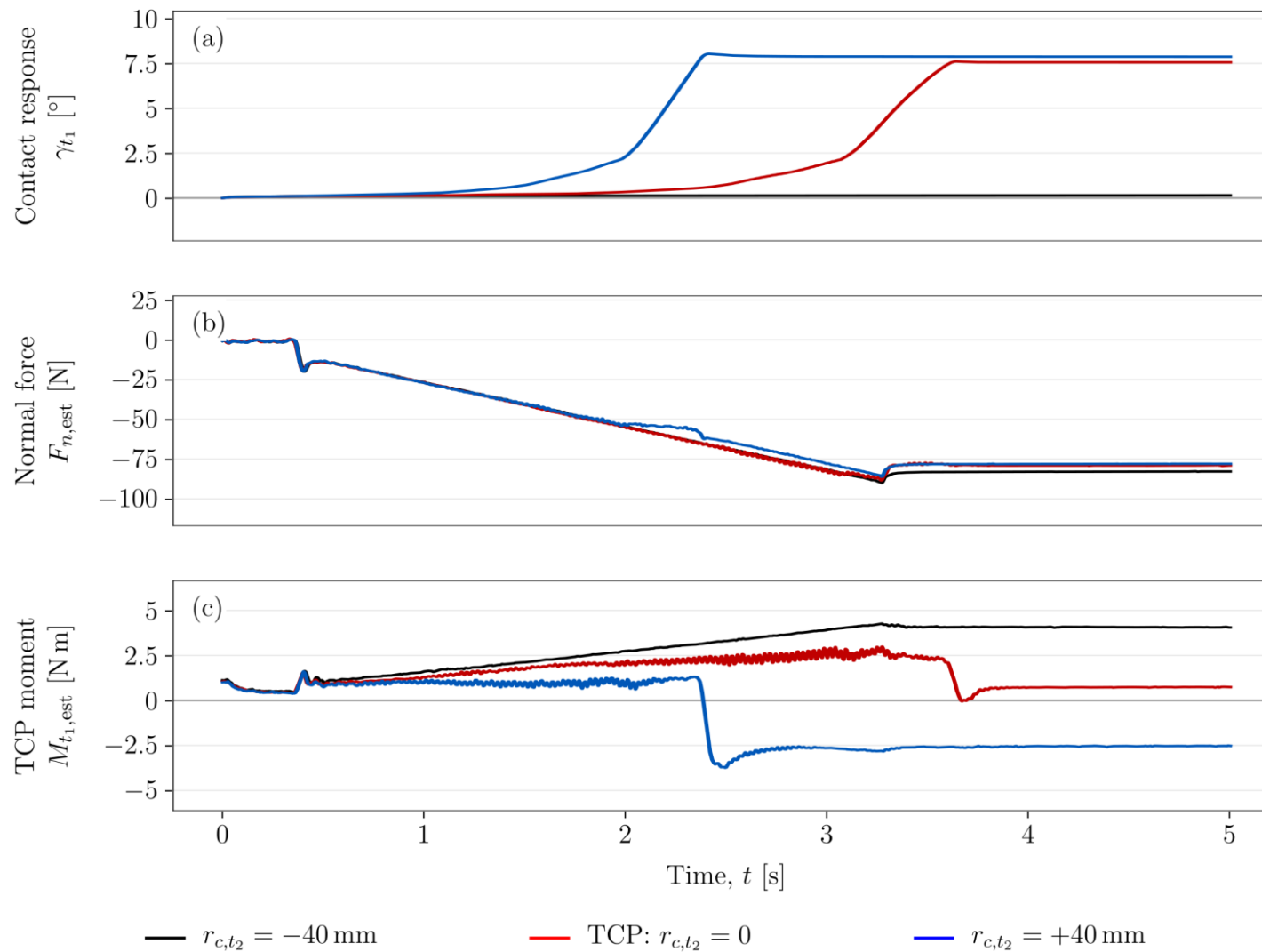
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_\sigma$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



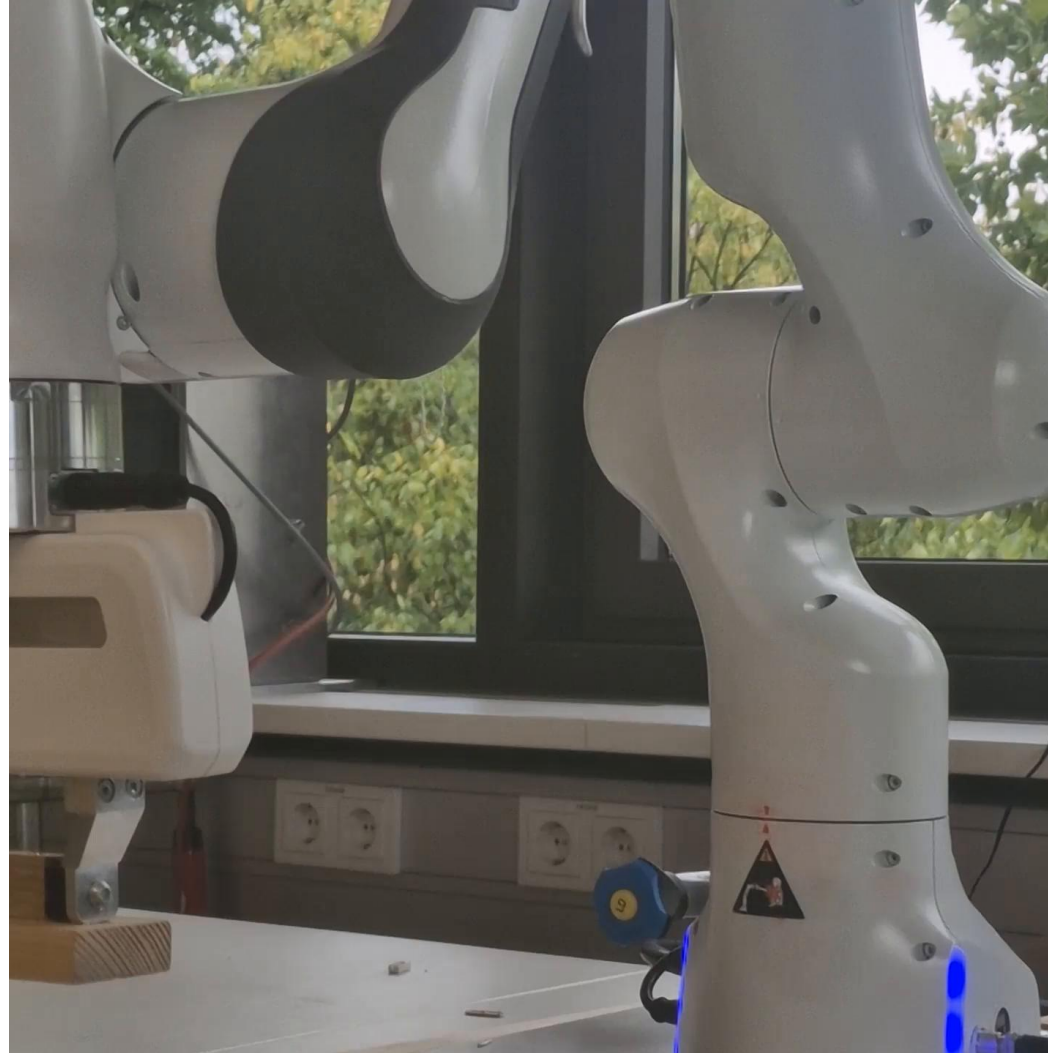
Representative traces
Positive entry tilt

- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates



Pre-grinding hold · fixed desired orientation · CoC at TCP

Disturbance demonstration



No null-space torque → Damping → Conditioning